



CS & IT ENGINEERING

Discrete Mathematics

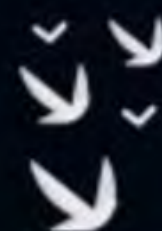
Graph Theory

Lecture No.- 04



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Recap of Previous Lecture



Topic

Degree Sequence



Topics to be Covered



Topic

Types of Graph Part-01



what will be value of x , where it is graphical?

$$0 \leq x \leq 5$$

$$x, 1, 2, 3, 5, 5$$

$$\overset{n-1}{5}, \overset{n-1}{5}, 3, 2, \textcircled{1}, x$$

Total vertices = 6

$$n-1, n-1, \dots, \textcircled{1}$$

it is not graphical.

it will never be graphical.

$$x = \text{N.A.}$$

Q.2: $x, 7, 7, \textcircled{5}, \textcircled{5}, 4, \textcircled{3}, 2$. $0 \leq x \leq 7$

Total vertices = 8

$$x \neq 1$$

According to Thm 2.

x will be only be odd.

$$\cancel{x}, \underline{3}, \underline{5}, 7$$

$$\text{Try } x = 3$$

$$x = 5$$

$$x = 7$$

Ans:
 $x = 3, 5$

Types of Graphs:

Complete Graph. (K_n) ($n \geq 1$)



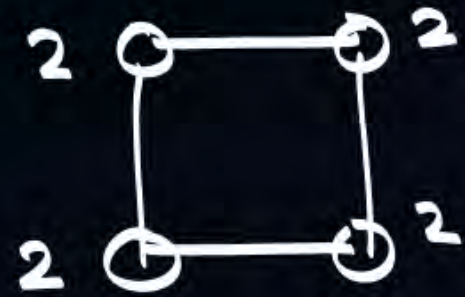
1. $\rightarrow$ it is having exactly $\frac{n(n-1)}{2}$ no. of edges.
2. $\rightarrow$ Degree of all vertices are $n-1$.

$$3. \left(\delta(G) = \frac{2e}{n} = \Delta(G) = n-1. \right)$$

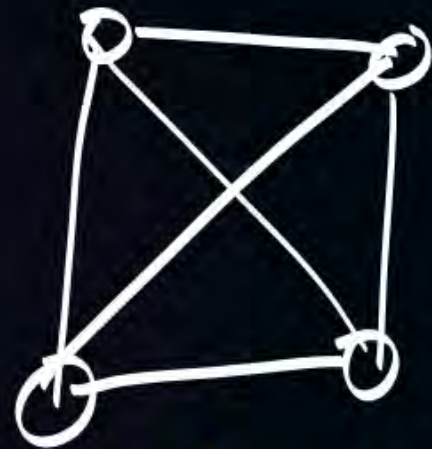
4. $n-1$ Regular.

Regular Graph. ($\delta(G) = \frac{2e}{n} = \Delta(G)$)

if Degrees of all vertices are same, then the graph is called Regular Graph.



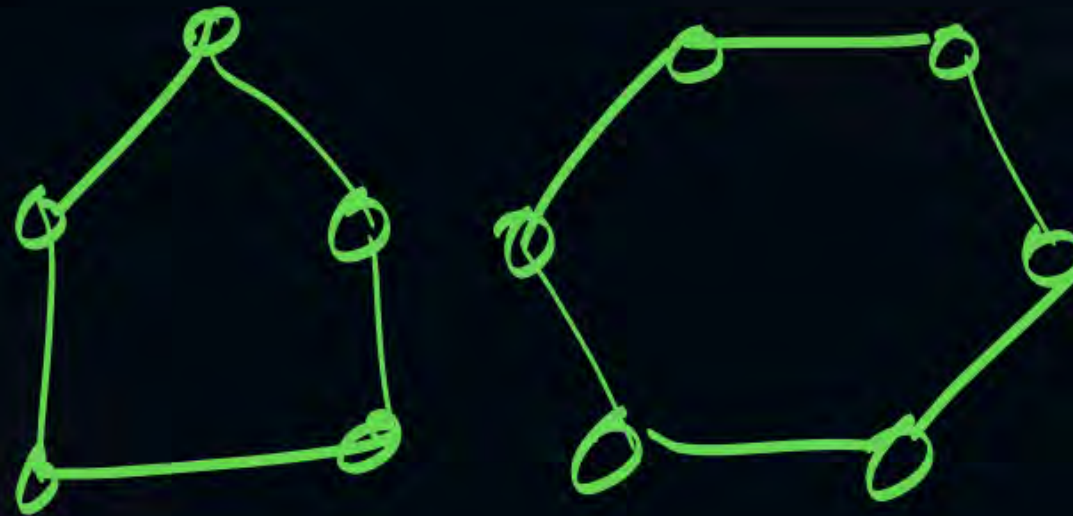
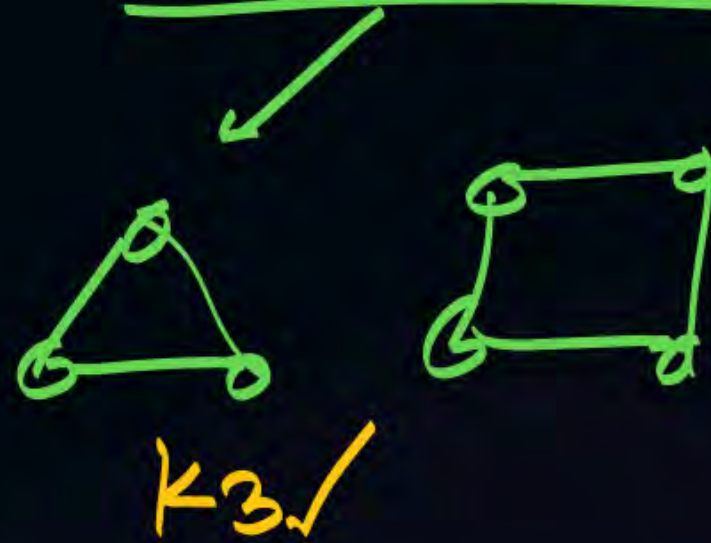
2-Regular Graph.



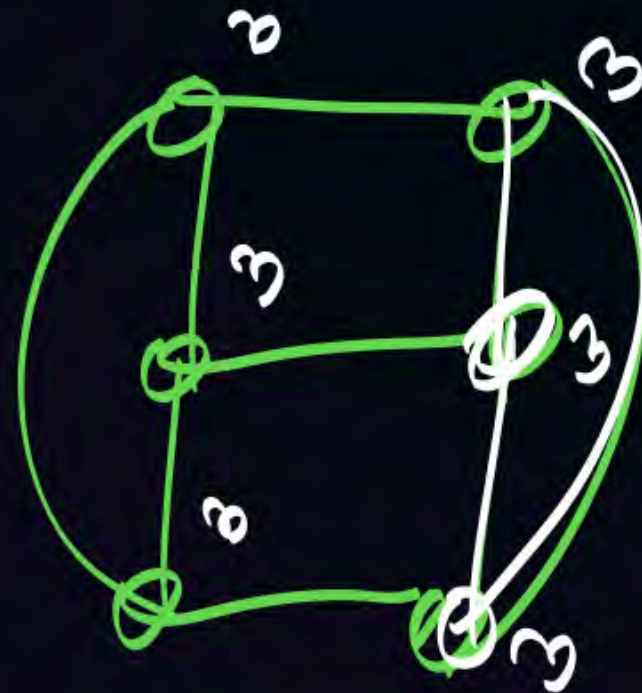
3-Regular Graph.

→ if degrees of all vertices are k , k -Regular Graph.

2-Regular



3-Regular



*

Every complete Graph
is Regular Graph.
but Every Regular Graph
need not be
Complete Graph.

Cycle Graph C_n ($n \geq 3$)

Even length cycle.

Odd length cycle.



C_3

$$\begin{cases} n=3 \\ e=3 \end{cases}$$



C_4

$$\begin{cases} n=4 \\ e=4 \end{cases}$$



C_5

$$\begin{cases} n=5 \\ e=5 \end{cases}$$

Degree of each vertex is 2.

→ 2-Regular Graph.

$$\left(\delta(G) = \frac{2e}{n} = \Delta(G) = 2 \right)$$

→ Every cycle Graph is Regular Graph, but viceversa is not True.

$$\sum d(v_i) = 2e.$$

$$n \cdot 2 = 2e$$

$$\downarrow \\ (n=e)$$

→ if Graph is C_n , then $n=e$ (True)

→ if Graph is having $(n=e)$ then G is cycle Graph. (false)

$$n = 5 \quad e = 5$$

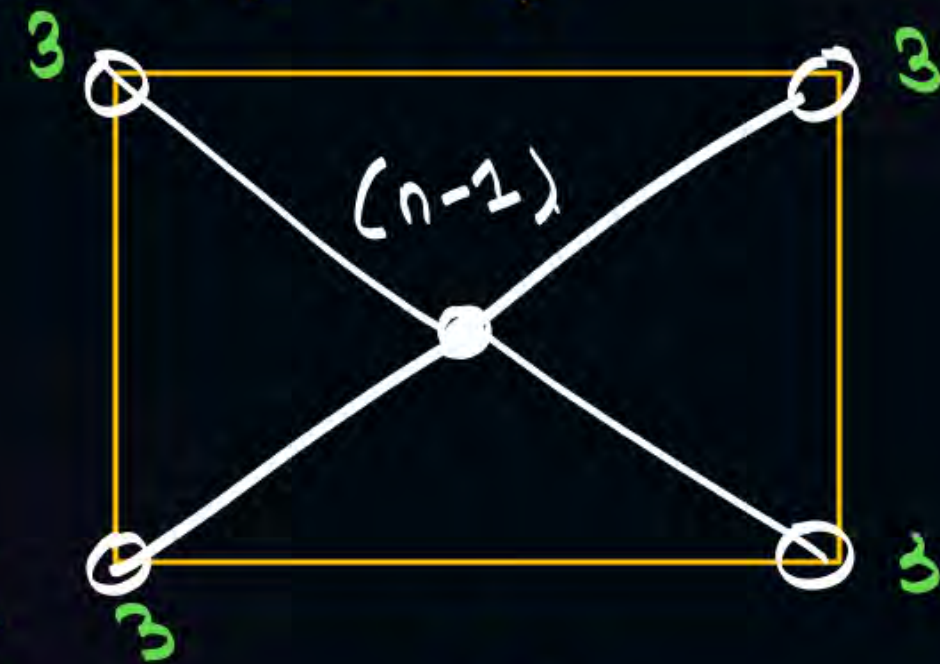
Graph which contains cycle (C_3)



not a Cycle Graph

↓
(Degrees of all vertices are 2)

Wheel Graph (W_n) ($n \geq 4$)



W_5

W_5



W_n

C_4

middle vertex.

$e=4$

$+ e=4$

$$e = n-1 + e = n-1$$

$$e(W_n) = 2(n-1)$$

W_{101}

$n=101$

C_{100}

$100, 3, \dots, 3$

$\times$

1 vertex, all remaining $(n-1)$

$\downarrow$
 $(n-1)$

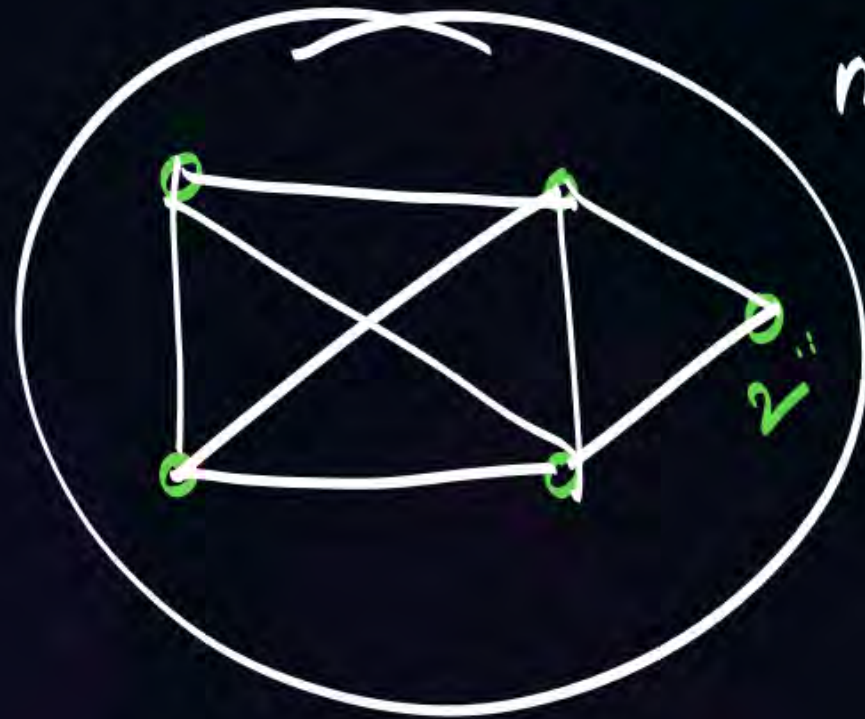
vertices will have degree 3.

if G is wheel Graph then $e = 2(n-1)$. (True)

if G is having $e = 2(n-1)$ then G is wheel Graph

$$n = 5 \quad e = 2(n-1)$$

$$= 2 \cdot 4 = 8$$



not wheel Graph.

Bipartite Graph:

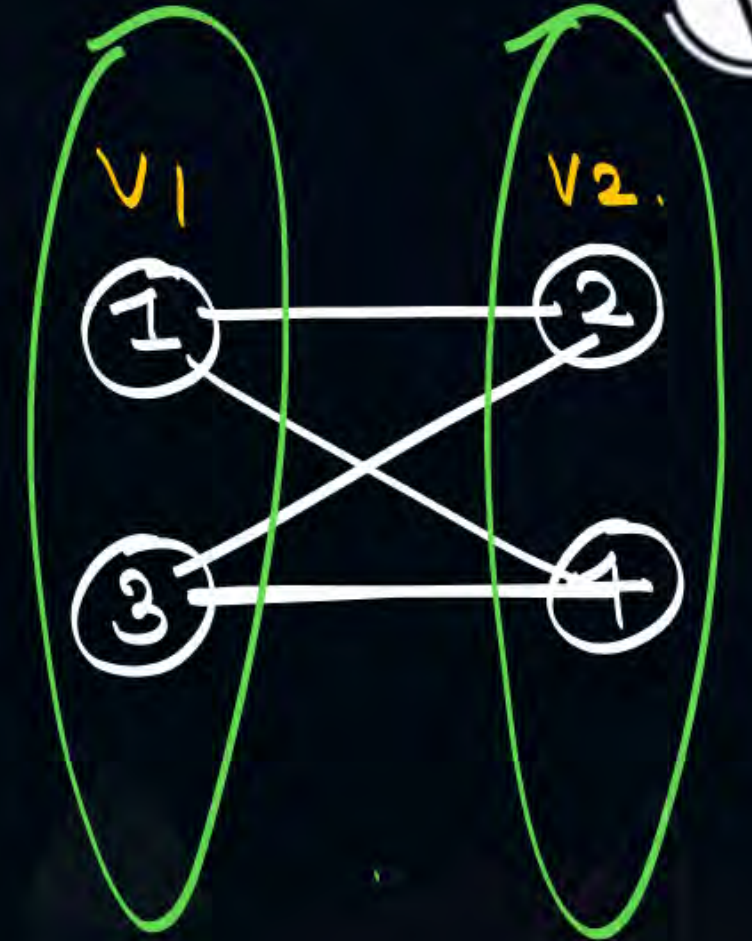
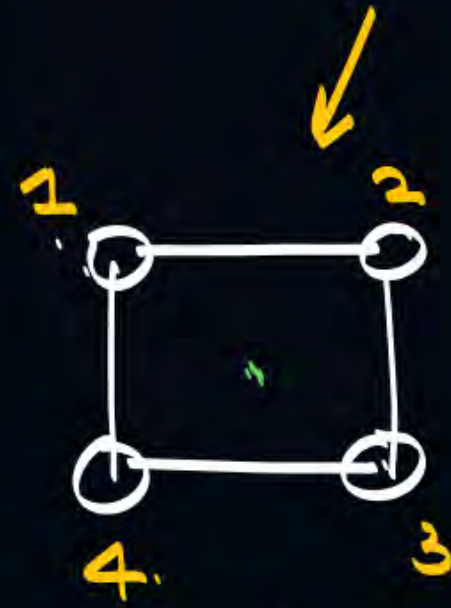
2.

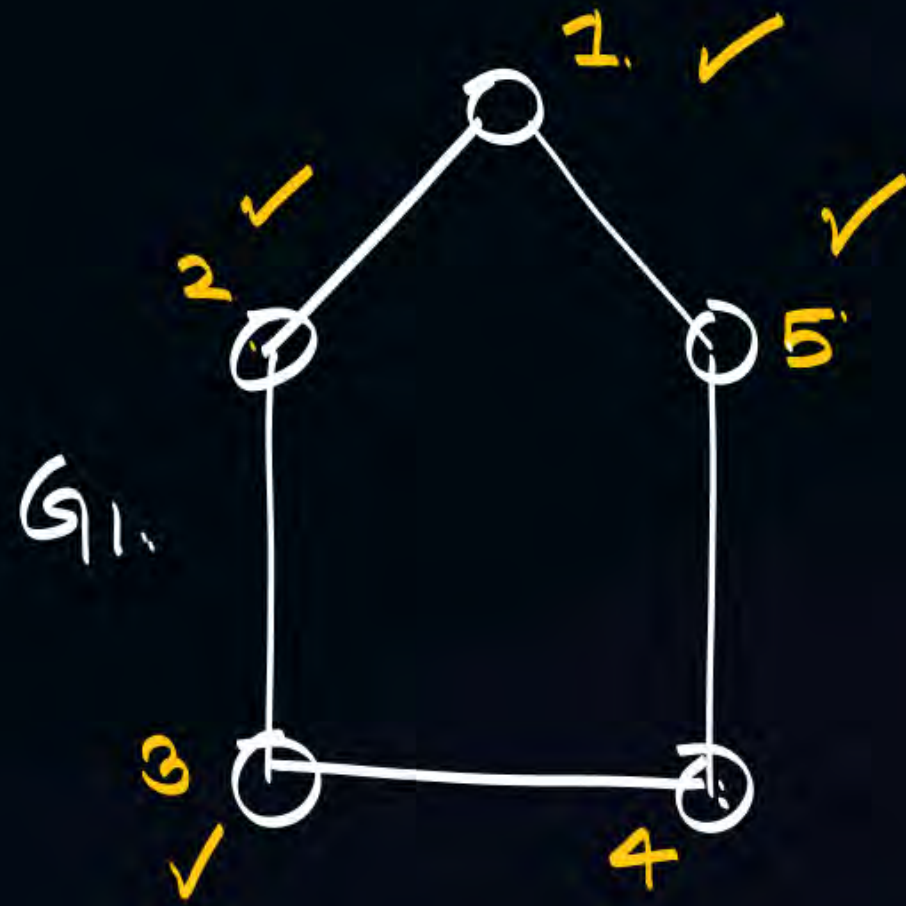
$$G = (V, E)$$



where V can be divided into 2-partitions such as V_1, V_2 .

all the edges, will be from one set to another set but not in a same set





it is not bipartite.
Graph.



not a
bipartite.
Graph.

Thm 6: Bipartite does not contains
odd length cycle.



G_1 is not B.P.G.



G_2 is also not B.P.G.

odd length cycle.



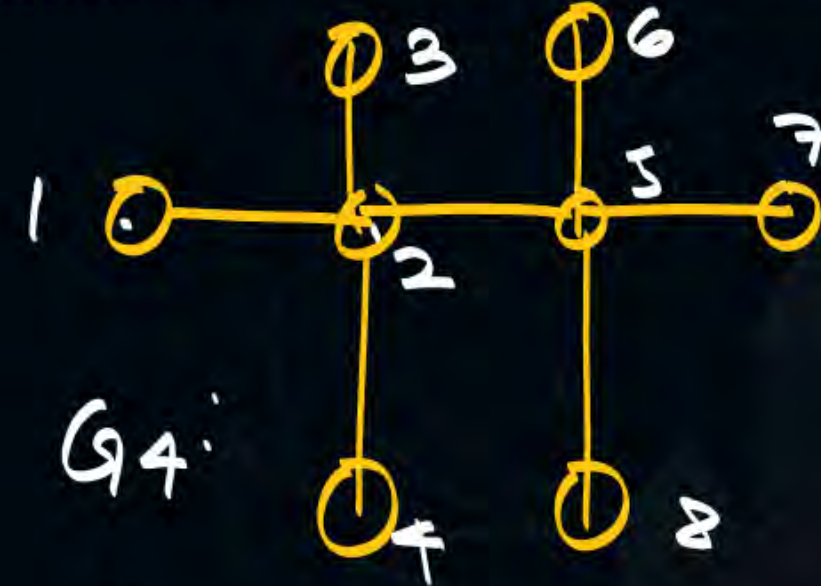
→ no odd length cycle
hence it is b.p.g.



no odd length cycle
Hence it is bipartite Graph.



→ G contains odd length cycle
Hence it is not bipartite Graph.



no cycle.
↓
it does not
contains odd
length cycle.
↓
Bipartite Graph





2 mins Summary

Topic

One

Types of Graph Part-01

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU